Contrasting Vertical Intergovernmental Interactions in Matching Fund Policy Implementation to Support Small and Medium-sized Enterprises in China

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This article uses evolutionary game theory to demonstrate a logically consistent explanation for the strategies of both superior and subordinate governments in different scenarios during the implementation of matching fund policy aimed at supporting the innovation activities of small and medium-sized enterprises. Through a case study of Shanghai, the article constructs a political interest–fiscal opportunity cost–manpower input cost model to explain the gaming behaviour of superior and subordinate governments. It further identifies a new form of project alienation in which subordinate governments conspire with enterprises to avoid providing matching funds and the causes of such alienation.

INTRODUCTION

Governmental support for the innovative activities of small and medium-sized enterprises (SMEs) is a global practice. As is evident in the large body of studies, there is an abundant interest in this topic,[1] with a focus on the efficiency and effectiveness of

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[1] Christos Dimos and Geoff Pugh, "The Effectiveness of R&D Subsidies: A Meta-Regression Analysis of the Evaluation Literature", *Research Policy* 45, no. 4 (2016): 797–815.

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governmental financial support to SMEs[2] and the causality[3] or correlativity of such support with the productivity of firm innovation.[4] However, an antecedent issue, namely the vertical intergovernmental interaction in the public subsidy process, has not received the attention it deserves. Admittedly, given the irreplaceable role of government in motivating the innovation of SMEs, considerable attention should be devoted to the governance processes and causal mechanisms within the government system related to subsidisation, especially since the topic of the relationships between government and market,[5] central and local governments,[6] local authorities and state principals[7] has long been a common interest among scholars. Embedded in these practical and academic backgrounds, this article seeks to shed new light on vertical intergovernmental interactions in the innovation support for SMEs in China by providing a clear window into this issue at the institutional level, namely the implementation of the so-called matching fund policy (MFP), which aims to coordinate the superior and subordinate governments in the subsidies of SME innovation activities.

The vertical relationship among governments is a classical issue in the field of public economics and governance[8] which focuses on the appropriate allocation of fiscal revenue and expenditure in the performance of government functions.[9] Since China is a centralised country characterised by an unitary political system, local leaders are not selected by voters, but are chosen by higher echelon leaders of the Communist

[2] Doh Soogwan and Kim Byungkyu, "Government Support for SME Innovations in the Regional Industries: The Case of Government Financial Support Program in South Korea", *Research Policy* 43, no. 9 (2014): 1557–69; Tor Jakob Klette, Jarle Møen and Zvi Griliches, "Do Subsidies to Commercial R&D Reduce Market Failures? Microeconometric Evaluation Studies", *Research Policy* 29, no. 4 (2000): 471–95.

[3] Josh Lerner, "The Government as Venture Capitalist: The Long-run Impact of the SBIR Program", *The Journal of Private Equity* 3, no. 2 (2000): 55–78.

[4] Maryann P. Feldman and Maryellen R. Kelley, "The Ex Ante Assessment of Knowledge Spillovers: Government R&D Policy, Economic Incentives and Private Firm Behavior", *Research Policy* 35, no. 10 (2006): 1509–21; Matthias Deschryvere, "R&D, Firm Growth and the Role of Innovation Persistence: An Analysis of Finnish SMEs and Large Firms", *Small Business Economics* 43, no. 4 (2014): 767–85.

[5] Andrew G. Walder, "Markets and Inequality in Transitional Economies: Toward Testable Theories", *American Journal of Sociology* 101, no. 4 (1996): 1060–73.

[6] Tim Conlan, "From Cooperative to Opportunistic Federalism: Reflections on the Half-Century Anniversary of the Commission on Intergovernmental Relations", *Public Administration Review* 66, no. 5 (2006): 663–76.

[7] Zhou Xueguang, "The Institutional Logic of Collusion Among Local Governments in China", *Modern China* 36, no. 1 (2010): 47–78.

[8] Herman L. Boschken, "Aligning a Multi-Government Network With Situational Context: Metropolitan Governance as an Organizational Systems Problem", *American Review of Public Administration* 47, no. 2 (2017): 189–208.

[9] B. Guy Peters and Jon Pierre, "Developments in Intergovernmental Relations: Towards Multi-Level Governance", *Policy & Politics* 29, no. 2 (2001): 131–5; Vincent Ostrom, Charles M. Tiebout and Robert Warren, "The Organization of Government in Metropolitan Areas: A Theoretical Inquiry", *American Political Science Review* 55, no. 4 (1961): 831–42; Liesbet Hooghe and Gary Marks, "Unraveling the Central State, But How? Types of Multi-level Governance", *American Political Science Review* 97, no. 2 (2003): 233–43.

Party;[10] its fiscal system functions as the "market-preserving federalism", essentially shaping the intergovernmental relationships at both horizontal and vertical levels. Intergovernmental relationship at the horizontal level, as an important incentive, leads to inter-jurisdictional competition,[11] while that at the vertical level, which is the research focus in this article, creates an imbalance of revenue powers and expenditure responsibilities within the vertical intergovernmental system.[12]

Innovation support, especially that for SMEs, is an indispensable function of government. In China, governments at all levels jointly shoulder responsibilities for promoting the innovation of SMEs by providing subsidies from the public budget to a variety of innovation projects. On closer scrutiny, the subsidisation process entails fundraising, funds distribution and funds management. At the outset, superior governments are supposed to assume a large part of the responsibilities to pay for projects that they have unilaterally decided to endorse. However, as a result of the mandate relationship,[13] subordinate governments gradually bear the burden of undertaking the matching fund responsibilities.[14] In other words, in addition to assisting in the preselection of projects, the corresponding subordinate government is required to provide matching funds for those subsidised projects in their jurisdictions. The process illustrates not only the origin of the MFP, but also vertical intergovernmental interaction.

The authors adopt evolutionary game theory analysis to demonstrate a logically consistent explanation of strategies of both superior and subordinate governments in different scenarios in MFP implementation, and also construct a political interest–fiscal opportunity cost–manpower input cost model using Shanghai as a case study to explain the gaming behaviour of superior and subordinate governments in MFP implementation. The authors have further identified a new form of project alienation in which subordinate governments conspire with enterprises to avoid providing matching funds and explain the causes of such alienation.

In outline, the contributions of this study are as follows. First, this study constructs an analytical framework based on evolutionary game theory for MFP implementation in the field of science and technology (S&T) innovation, and portrays the actual interaction between vertical governments in the project system. This framework can

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[10] Fan Yongmao, "The Centre Decides and the Local Pays: Mandates and Politics in Local Government Financial Management in China", *Local Government Studies* 41, no. 4 (2015): 516–33.

[11] Geoffrey Brennan and James M. Buchanan, *The Power to Tax: Analytical Foundations of a Fiscal Constitution* (Cambridge: Cambridge University Press, 1980).

[12] Loraine A. West and Christine P.W. Wong, "Fiscal Decentralization and Growing Regional Disparities in Rural China: Some Evidence in the Provision of Social Services", *Oxford Review of Economic Policy* 11, no. 4 (1995): 70–84; Jorge Martinez-Vazquez and Robert M. McNab, "Fiscal Decentralization and Economic Growth", *World Development* 31, no. 9 (2003): 1597–616.

[13] Paul L. Posner, *The Politics of Unfunded Mandates: Whither Federalism?* (Washington, DC: Georgetown University Press, 1998).

[14] Jin Hehui, Qian Yingyi and Barry R. Weingast, "Regional Decentralization and Fiscal Incentives: Federalism, Chinese Style", *Journal of Public Economics* 89, no. 9 (2005): 1719–42.

be extended to a general study of complementary policies and is instructive for other countries in gaining an understanding of the relationship between different levels of government in implementing a project system. Second, this article highlights the dual economic and social attributes of MFP policies and identifies factors such as political interest, fiscal opportunity cost and manpower input cost as playing important roles in vertical intergovernmental interactions. It also emphasises the significance of policy fields underlying vertical intergovernmental relations in the study of issues such as project-based and vertical intergovernmental relations. The types of benefits and costs perceived by superior and subordinate governments and their relative importance vary across fields constituted by policies with different attributes, and the combination of different benefits and costs leads to differences in their dynamic relationships. Third, the article accentuates the importance of governance structure in the allocation of fiscal resources for science, technology and innovation. Regulations regarding the proportion of matching contributions from the superior to the subordinate government, penalties for non-compliance, and rewards for compliance can all lead to different behavioural responses by governance actors, which in turn affect the efficiency of the external allocation of funds.

This article follows closely Fan Yongmao's study, which discusses local governments' divergent responses and coping strategies when they have to deal with payments for education, public health care and affordable housing construction, etc.[15] By applying comprehensive theoretical analysis and case study methods, this article uniquely examines the interactive behaviour between vertical governments in the field of the MFP and reformulates this behaviour as "the superior government decides and pays, and the subordinate government cooperates in the payment". Specifically, the authors aim to establish the existence of vertical intergovernmental interaction in the implementation of the MFP, focusing on the strategies of both superior and subordinate governments as shaped by various factors.

The article is structured as follows. The second section offers a brief background of matching fund policy. The third section details the application of evolutionary game theory to demonstrate a logically consistent explanation of the evolutionary stable strategies (ESS) of both superior and subordinate governments in different scenarios of the MFP implementation. The fourth section introduces the case materials and the data collection procedures, and summarises the interview findings of various vertical intergovernmental interactions in the implementation of the MFP. The fifth section explains the findings using theoretical propositions. In the sixth section, the authors conclude and offer some recommendations.

[15] Fan, "The Centre Decides and the Local Pays".

THE INSTITUTIONAL AND FACTUAL BACKGROUNDS

"Project System" and the Rise of Matching Fund Policy (MFP)

The "project system" is the outcome of the tax-sharing reform and has been the mainstream form of fiscal resource allocation during the past decade in China, involving vertical intergovernmental interactions both on matters of fiscal revenue and expenditure and on the division of responsibilities. In the context of this article, the "project system" in the field of innovation support is reflected in the direct subsidies to selected SMEs in the form of proposed innovation projects. The management of projects follows a systematic procedure of "application, preliminary examination, formal approval, execution and supervision, inspection and evaluation, reward or punishment". Source of funding and task assignment are two key elements in project management.

Within the framework in which different levels of government interact with reference to innovation support for SMEs, the superior government occupies an absolute dominant position in the approval of proposed innovation projects; thus the superior government should be the payer of the promised funds to those approved SMEs. However, in practice, the trend mentioned above, that "the superior government decides and pays, and the subordinate government cooperates in the payment", means that the duty of raising funds is jointly performed by governments of different levels that include both superior and subordinate governments, and more specifically, the subordinate government is required to provide matching funds for the approved SMEs geographically registered in its jurisdiction at a fixed or flexible rate. Furthermore, the subordinate government is also required to undertake daily supervision of these SMEs. Above all, the matching fund policy suggests a strategy adopted by a superior government in order to deal with the source of funding and assignment of tasks in the innovation support for SMEs.

Essentially, there are three explanations why the superior government propels the implementation of the MFP. First, the superior government asserts that the subordinate governments should subsidise SMEs in their respective jurisdictions, as they are the direct beneficiaries of technology or economic spillover from the business growth. Second, the superior government aims to play a leverage role in facilitating SMEs' synchronous financial support from the corresponding subordinate governments, reducing its own fiscal burden and expanding the scope of funding benefits. Third, the superior government aims to capitalise on subordinate governments' enthusiasm to support SMEs as the subordinate governments have a local information advantage in efficiently and effectively supervising innovation projects.

International Background and MFP Implementation in China

From a global perspective, China's matching fund policy may be compared to the grants-in-aid which were originally carried out by the US Congress in 1835, upon paying off the national debt and forecasting future budget surpluses indefinitely, to distribute leftover funds to the states on the basis of population for the purpose of

financing public works.[16] Intergovernmental funds transferred from the federal government have since been used to support state and local programmes, while state and local governments are evidently responsible for much of the increase in expenditures as well as possible budget surplus.[17] China's MFP, in fact, shares a greater similarity with the grant programmes of the Federal Transit Administration for transit agencies, which require either the state or the local entity to contribute a certain percentage of funds to match the federal grant.[18]

Today, local governments of most regions in China have adopted the MFP. However, there are regional differences in policy details, in terms of setting stringent instructions, the matching ratio and the upper limit of matching fund amount. Matching ratio refers to the proportion of funding provided by the subordinate government to that provided by the superior government. For example, when the superior government contributes one million yuan, the subordinate government would have to contribute 300,000 yuan if a 30 per cent matching ratio is applied. Generally, local governments in more developed regions are more likely to adopt a higher matching ratio. For example, in Shenzhen, Longgang and Futian districts adopt a matching ratio at 50 per cent, compared to the less-developed Yantian district, which adopts a lower but flexible ratio of between 30 and 50 per cent. Moreover, depending on whether the project is approved by the central government or by the provincial government, the subordinate government usually takes different supporting measures. For instance, the matching ratio for central government-approved projects always ranges from 80 to 100 per cent with an upper limit of one million yuan and above, while that for provincial government-approved projects varies flexibly from 30 to 70 per cent with an upper limit of less than several hundred thousand yuan.

However, with MFP implementation, several issues, of which project alienation is more salient, began to emerge. The literature identifies two main types of alienation. One is unconscious implementation bias caused by objective factors;[19] the other is grassroots governments' deliberate policy avoidance to evade responsibility or distort the policies for profit. This is manifested mainly in false project declarations and project imitation.[20] Most studies analyse the reasons for project alienation from the perspective of "control rights". Zhou introduces the theory of "control rights" to explain the diversity of project implementation outcomes and argues that the distribution

[16] James R. Hines and Richard H. Thaler, "Anomalies: The Flypaper Effect", *The Journal of Economic Perspectives* 9, no. 4 (1995): 217–26.

[17] Edward M. Gramlich and Harvey Galper, "State and Local Fiscal Behavior and Federal Grant Policy", *Brookings Papers on Economic Activity* 4, no. 1 (1973): 15–66.

[18] National Academies of Sciences, Engineering, and Medicine, *Identification of Local Matching Fund Requirements for State-Administered Federal and Non-Federal Public Transportation Programs* (Washington, DC: The National Academies Press, 2011).

[19] Yang Bao, "Embedded Structure, Resource Mobilization and Contract Performance" (in Chinese), *Journal of Public Management* 15, no. 3 (2018): 39–50.

[20] Wu Kechang and Ou Wanyi, "Imitation of Projects: An Alienated Form of Grassroots Project-based Operation" (in Chinese), *Studies on Party and Government*, no. 4 (2021): 21–9.

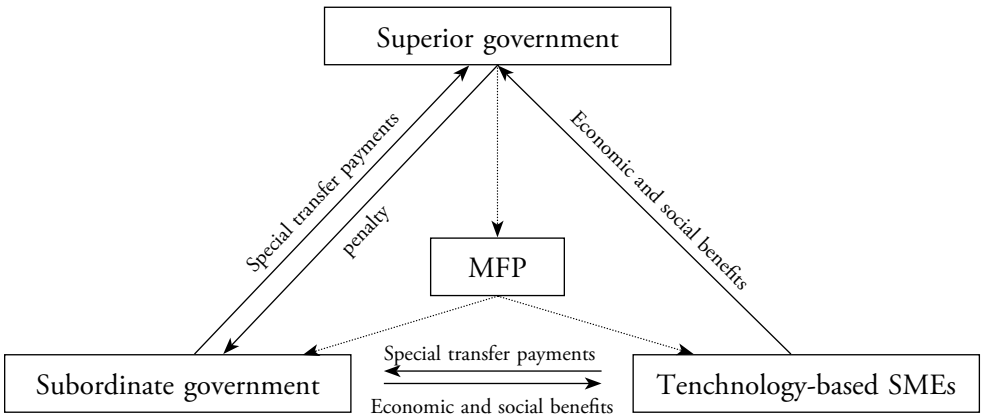
and combinations of control between the central and local government have led to different approaches of project system implementation.[21] There are three dimensions of control, namely goal-setting, inspection and acceptance, and implementation/incentives. Most of the literature analyses the causes of project alienation from specific research cases and identifies three main types of influencing factors that can be summarised as the mismatch between project objectives and development goals, the pursuit of fiscal revenue and uneven distribution of the local fiscal base.

THEORETICAL ANALYSIS

The Evolutionary Game Theory Framework

Evolutionary game theory, first proposed by Smith and Price,[22] provides a useful approach to describing the evolutionary dynamics of a well-mixed population. Based on the bounded rationality hypothesis, the theory is effective for analysing competition when two players can cooperate or defect in each round.[23] Weighing the losses and gains, the superior–subordinate government relationship can be considered a gaming situation (Figure 1). Hence, the authors, drawing on evolutionary game theory, aim to derive a better understanding of the intergovernmental interactions and the coordinating scheme to enforce the MFP.

Figure 1. Gaming Situation Between Superior and Subordinate Governments



Notes: MFP denotes matching fund policy; SMEs denotes small- and medium-sized enterprises.

[21] Zhou Xueguang, "The Project System: A Perspective of the 'Controlling Power' Theory" (in Chinese), *Open Times*, no. 2 (2015): 82–102.
[22] J. Maynard Smith and G.R. Price, "The Logic of Animal Conflict", *Nature* 246 (1973): 15–8.
[23] Robert Axelrod, *The Evolution of Cooperation* (New York: Basic Books, 1984).

Assumptions

The authors formulate the following basic hypotheses in order to simplify the analysis:

- Two players: the superior government and subordinate government are respectively denoted as U and L; both have bounded rationality.
- The strategy set for the superior government is Au={supervise, not supervise}, while that for the subordinate government is Al={enthusiastic engagement, indolent engagement}. "Enthusiastic engagement" means that the subordinate government implements the policy requirements of the superior government in a step-by-step manner and provides matching funds to companies approved by the superior government as required. "Indolent engagement" means that the subordinate government does not carry out the requirements of the superior government, or that it finds ways to pretend that it has done so, e.g. by falsifying documents or colluding with companies.
- When the subordinate government enthusiastically engages in the MFP, it would monitor the corporations more carefully than when it indolently engages in the MFP in order to prevent its own funds from being compromised.
- As long as the superior government supervises, it can judge the behaviour of the subordinate government and will punish the latter for indolent engagement.
- The enthusiastic engagement of the subordinate government regarding the MFP will produce a spillover effect on the superior government's benefits.

Payoff matrix

In order to simulate the behaviour of the superior government and the subordinate government in the implementation process of the MFP, the payoff functions and impact factors of the two players are analysed. The parameters used in the payoff functions are denoted as follows:

- C_u: the supervision cost of the superior government.
- C_{le}: the engagement cost of the subordinate government for enthusiastic engagement, including fiscal opportunity cost, manpower input cost and risk event cost.
- C_{li}: the engagement cost of the subordinate government for indolent engagement.
- P: the punishment exacted from the subordinate government by the superior government due to indolent engagement. This takes the form of complementary funding.
- B_u: the benefit for the superior government from innovation support for SMEs.
- B_{le}: the benefit for the subordinate government from enthusiastically engaging in the MFP, including economic, social and political benefits.
- B_{li}: the benefit for the subordinate government from indolently engaging in the MFP.
- α: the rate of spillover benefit for the superior government when the subordinate government enthusiastically engages in the MFP; its value ranges from 0 to 1.

Table 1 demonstrates the payoff matrix of the superior and subordinate governments denoted by the aforementioned parameters.

TABLE 1
PAYOFF MATRIX OF THE SUPERIOR AND SUBORDINATE GOVERNMENTS

		Subordinate Government	
		Enthusiastic engagement	Indolent engagement
Superior Government	Supervise	$B_u+\alpha B_{le}-C_u$, $B_{le}-C_{le}$	B_u+P-C_u, $B_{li}-C_{li}-P$
	Not Supervise	$B_u+\alpha B_{le}$, $B_{le}-C_{le}$	B_u, $B_{li}-C_{li}$

Replication Dynamic Mechanism and Evolutionary Stable Strategies

Assumed in the MFP implementation process is that the proportion of enthusiastic subordinate governments is x and the proportion of indolent subordinate governments is (1–x). Also assumed is that the proportion of superior governments that choose to supervise or not to supervise is, respectively, y and (1–y).

For the subordinate government

The expected return for enthusiastically engaging in MFP, U_{le}, is:

$$U_{le} = y(B_{le} - C_{le}) + (1-y)(B_{le} - C_{le}) = B_{le} - C_{le} \tag{1}$$

The expected return for indolently engaging in MFP, U_{li}, is:

$$U_{li} = y(B_{li} - C_{li} - P) + (1-y)(B_{li} - C_{li}) = B_{li} - C_{li} - yP \tag{2}$$

The average return of the subordinate government is:

$$\overline{U}_l = xU_{le} + (1-x)U_{li} \tag{3}$$

The replication dynamics equation of the subordinate government is therefore:

$$F(x) = \frac{dx}{dt} = x(U_{le} - \overline{U}_l) = x(1-x)(U_{le} - U_{li})$$
$$= x(1-x)(yP + B_{le} - C_{le} - B_{li} + C_{li}) \tag{4}$$

For the superior government

The expected return for supervising is:

$$U_{uy} = x(B_u + \alpha B_{le} - C_u) + (1-x)(B_u + P - C_u)$$
$$= B_u + P - C_u + x\alpha B_{le} - xP \tag{5}$$

The expected return for not supervising is:

$$U_{un} = x(B_u + \alpha B_{le}) + (1-x)B_u = B_u + x\alpha B_{le} \tag{6}$$

The average return of the subordinate government is:

$$\overline{U}_u = yU_{uy} + (1-y)U_{un} \tag{7}$$

The replication dynamics equation of the superior government is therefore:

$$G(y) = \frac{dy}{dt} = y(U_{uy} - \overline{U}_u) = y(1-y)(U_{uy} - U_{un})$$
$$= y(1-y)(P - C_u - xP) \tag{8}$$

With regard to equations (4) and (8), they can be expressed as a two-dimensional dynamic autonomy system where

$$\begin{cases} F(x) = x(1-x)(yP + B_{le} - C_{le} - B_{li} + C_{li}) \\ G(y) = y(1-y)(P - C_u - xP) \end{cases} \tag{9}$$

According to the two-dimensional differential theorem, there are five equilibrium points within the system: O(0, 0), A(1, 0), B(1, 1), C(0, 1) and D(x*, y*), where $x^*=(P-C_u)/P$, $y^*=(B_{li}-C_{li}-B_{le}+C_{le})/P$, and both x* and y* should range from 0 to 1.

To solve the partial derivative of F(x), G(y) with respect to x and y respectively, the authors use the Jacobian matrix of equation (9):

$$J = \begin{bmatrix} \dfrac{\partial F(x)}{\partial x} & \dfrac{\partial F(x)}{\partial y} \\ \dfrac{\partial G(y)}{\partial x} & \dfrac{\partial G(y)}{\partial y} \end{bmatrix}$$
$$= \begin{bmatrix} (1-2x)(yP + B_{le} - C_{le} - B_{li} + C_{li}) & P(1-x)x \\ P(y-1)y & (1-2y)(P - C_u - xP) \end{bmatrix}$$

The determinant of J is:

$$\det J = (1-2x)(yP + B_{le} - C_{le} - B_{li} + C_{li})(1-2y)(P - C_u - xP)$$
$$- P(1-x)xP(y-1)y$$

And its trace is:

$$trJ = (1-2x)(yP + B_{le} - C_{le} - B_{li} + C_{li}) + (1-2y)(P - C_u - xP)$$

According to the methods for computing evolutionary stable strategy (ESS) of the system proposed by Friedman,[24] two conditions should be simultaneously met to recognise an equilibrium point where the ESS exists: detJ>0 and *tr*J<0; if the symbol of *tr*J is uncertain, the equilibrium point is a saddle point, otherwise the equilibrium point is an unstable point.

The determinant of J and its trace for each equilibrium point are shown in Table 2.

TABLE 2
EQUILIBRIUM POINT AND DETJ AND TRJ VALUES

Point	*detJ*	*trJ*
O(0, 0)	$(B_{le}-C_{le}-B_{li}+C_{li})(P-C_u)$	$(B_{le}-C_{le}-B_{li}+C_{li})+(P-C_u)$
A(1, 0)	$(B_{le}-C_{le}-B_{li}+C_{li})C_u$	$-(B_{le}-C_{le}-B_{li}+C_{li})-C_u$
B(1, 1)	$(P+B_{le}-C_{le}-B_{li}+C_{li})(-C_u)$	$-(B_{le}-C_{le}-B_{li}+C_{li})-(P-C_u)$
C(0, 1)	$-(P+B_{le}-C_{le}-B_{li}+C_{li})(P-C_u)$	$B_{le}-C_{le}-B_{li}+C_{li}+C_u$
D(x*,y*)	$\frac{C_u(P+B_{le}-C_{le}-B_{li}+C_{li})(P-C_u)(B_{li}-C_{li}-B_{le}+C_{le})}{P^2}$	0

Distinguishing six different scenarios, the authors recognise the existence of local stability of the system at each equilibrium point in the corresponding scenario by respectively judging the symbol of detJ and that of *tr*J. Table 3 outlines the findings.

TABLE 3
STABILITY ANALYSIS OF THE EVOLUTIONARY GAME MODEL EQUILIBRIUM POINT

	Scenario 1: $B_{le}-C_{le}-B_{li}+C_{li}>0$, $P-C_u>0$			**Scenario 2:** $B_{le}-C_{le}-B_{li}+C_{li}>0$, $P-C_u<0$			**Scenario 3:** $B_{le}-C_{le}-B_{li}+C_{li}<0$, $P-C_u<0$, $P+B_{le}-C_{le}-B_{li}+C_{li}<0$		
Equilibrium point	*Symbol of detJ*	*Symbol of trJ*	*Local stability*	*Symbol of detJ*	*Symbol of trJ*	*Local stability*	*Symbol of detJ*	*Symbol of trJ*	*Local stability*
O(0, 0)	+	+	unstable	–	±	saddle	+	–	**ESS**
A(1, 0)	+	–	**ESS**	+	–	**ESS**	–	±	saddle
B(1, 1)	–	–	unstable	–	±	saddle	+	+	unstable
C(0, 1)	–	+	unstable	+	+	unstable	–	±	saddle
D(x*,y*)	Not applicable			Not applicable			Not applicable		

(*Cont'd overleaf*)

[24] Daniel Friedman, "Evolutionary Games in Economics", *Econometrica* 59, no. 3 (1991): 637–66.

TABLE 3 (*Cont'd*)

	Scenario 4: $B_{le}-C_{le}-B_{li}+C_{li}<0$, $P-C_u<0$, $P+B_{le}-C_{le}-B_{li}+C_{li}>0$			**Scenario 5:** $B_{le}-C_{le}-B_{li}+C_{li}<0$, $P-C_u>0$, $P+B_{le}-C_{le}-B_{li}+C_{li}<0$			**Scenario 6:** $B_{le}-C_{le}-B_{li}+C_{li}<0$, $P-C_u>0$, $P+B_{le}-C_{le}-B_{li}+C_{li}>0$		
Equilibrium point	*Symbol of detJ*	*Symbol of trJ*	*Local stability*	*Symbol of detJ*	*Symbol of trJ*	*Local stability*	*Symbol of detJ*	*Symbol of trJ*	*Local stability*
O(0, 0)	+	–	**ESS**	–	–	unstable	–	±	saddle
A(1, 0)	–	–	unstable	–	+	unstable	–	±	saddle
B(1, 1)	–	+	unstable	+	+	unstable	–	±	saddle
C(0, 1)	+	+	unstable	+	–	**ESS**	–	±	saddle
D(x*,y*)	Not applicable			Not applicable			+	0	unstable

Notes: ESS denotes evolutionary stability strategy.

In summary, results in scenarios 1 and 2 show that where the net benefit to the subordinate government from enthusiastically engaging in the MFP (B_{le}–C_{le}) is higher than that from indolently engaging in the MFP (B_{li}–C_{li}), the evolutionary stable strategies for the subordinate government and superior government are "enthusiastic engagement" and "not supervise", respectively, regardless of whether the punishment by the superior government is higher than its supervision cost.

Results in scenarios 3 and 4 show that where the net benefit to the subordinate government from enthusiastically engaging in the MFP is less than that from indolently engaging in the MFP and the punishment by the superior government is less than its supervision cost, the evolutionary stable strategies for the subordinate government and superior government are "indolent engagement" and "not supervise", respectively.

The result in scenario 5 shows that where the net benefit to the subordinate government from enthusiastically engaging in the MFP is less than that from indolently engaging in the MFP, the punishment by the superior government is more than its supervision cost and the sum of punishment by the superior government and the net benefit to the subordinate government relative to enthusiastic engagement and indolent engagement respectively are negative, the evolutionary stable strategies for the subordinate government and superior government are "indolent engagement" and "supervise", respectively.

The result in scenario 6 shows that where the net benefit to the subordinate government from enthusiastically engaging in the MFP is less than that from indolently engaging in the MFP, the punishment by the superior government is more than its supervision cost, and the sum of punishment by the superior government and the net benefit to the subordinate government relative to enthusiastic engagement and indolent engagement respectively are positive, there is no evolutionary stable strategy for the subordinate government and superior government.

Theoretical Propositions

In reference to the above analysis, the authors put forward the following theoretical propositions:

Proposition 1

The benefit and cost of engaging in the MFP and the punishment by superior government due to indolent engagement are factors that impact the engagement behaviour of the subordinate government, whereas the supervision cost, the engagement behaviour of the subordinate government and the punishment given to the subordinate government due to indolent engagement are factors that impact the behaviour of the superior government.

Proposition 2

For the subordinate government, when the net benefit from enthusiastically engaging in the MFP is greater than that from indolently engaging in MFP, it would always be enthusiastic. Otherwise, only if the punishment is more than the superior government's supervision cost, the sum of punishment and the relative net benefit to the subordinate government are positive and the superior government strengthens supervision at the same time, would it be likely to be enthusiastic; in other circumstances, it would be indolent.

Proposition 3

For the superior government, only if the punishment is more than the supervision cost and the net benefit to the subordinate government from enthusiastically engaging in the MFP is less than that from indolently engaging in MFP, would it supervise the subordinate government; otherwise, it would not supervise the latter.

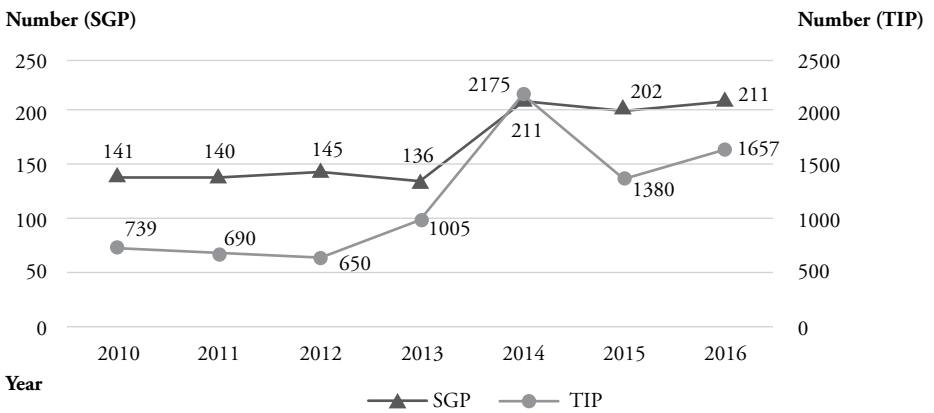
METHODOLOGICAL APPROACH

Case Selection

The authors' research team organised a group of five researchers to undertake an investigation in Shanghai city in 2017. As one of the four municipalities in China, Shanghai is the country's economic, financial, trade and shipping centre, and has always led the nation's innovation trend. The vigorous development of various SMEs has played an irreplaceable role in advancing the city's innovative capacity. In May 2017, local leaders announced their ambition to establish Shanghai as the scientific and technological innovation centre of global influence, and implemented strategic measures to reform technology fund management as an important part of the announced strategic measures.

There are two programmes related to the innovation support for SMEs initiated by the Municipal Government of Shanghai: the "small giant programme" (SGP) and the "technological innovation programme" (TIP). These directly support SMEs at the start-up stage and at the development stage, respectively. For SMEs that have been approved by the SGP or TIP, the municipal government will provide a direct grant for subsidising innovation activities and will regulate the local district government's provision of matching funds in a ratio of 1:1. Figure 2 shows the number of projects approved by the SGP and TIP from year to year.

Figure 2. Number of Projects Approved by the SGP and TIP



Notes: SGP denotes small giant programme; TIP denotes "technological innovation programme".
Source: Data set collected by the authors through field research.

Shanghai city currently has 16 districts under its jurisdiction; these districts differ significantly in endowments such as economic development level, governmental fiscal capacity, industrial development priorities, attractiveness to innovative enterprises, etc. Given that the municipal government as a superior government has put forward the unified policy requirement of providing matching funds, a comprehensive picture of vertical intergovernmental interactions in the implementation of the MFP can be delineated through investigating the engagement behaviour of different representative district governments in a contrasting perspective.

Based on the principles of representativeness, difference and convenience, the authors select three districts in which to conduct the field investigation. A brief introduction for each case district is as follows. Table 4 provides basic parameters of the three districts in 2017 in more specific detail.

- ***P district***

  P district has the largest area, the largest population, the most abundant fiscal resources and the most advanced high-tech industry in Shanghai. The most famous high-tech park in P district is called a Silicon Valley of China.

- ***C district***
  C district consists of three islands; its development orientation is as the ecological barrier of as well as the ecological strategic space for the future of Shanghai; its leading industries are modern agriculture and tourism.
- ***Y district***
  Y district is not only an old industrial base of Shanghai, it also has a large number of research institutes and universities. Since the 1990s it has been dedicated to transforming itself into a modern innovation base leveraging on its scientific and educational advantages, and it thus has a higher budget in technological innovation expenditure.

TABLE 4
Basic Parameters of the Three Districts in 2017

Basic parameters	GDP (billion yuan)	Fiscal revenue (billion yuan)	Number of SMEs	Fiscal expenditure on S&T (billion yuan)	Number of SGPs
P district	965.14	393.80	247,185	39.08	21
C district	70.75	41.81	187,138	0.53	4
Y district	170.32	95.34	50,446	1.49	4

Notes: GDP denotes gross domestic product; SMEs, small- and medium-sized enterprises; SGP, small giant programme.

Data Collection

The authors' investigation of Shanghai city integrated two primary approaches to data collection. The first is documentary research and analysis of policy texts relating to the MFP when implementing the SGP and TIP. The second involves field interviews with officials and staff familiar with the MFP implementation in the competent departments of the case districts.

For the documentary research, the authors used a search-oriented procedure to identify the documents associated with the MFP in operating SGP and TIP that were drafted between 2010 and 2016. Providing insight into core contents of the MFP, these documents are organised into two broad categories. The first comprises formal policy documents that provide texts on the regulations of the MFP, including the "measures" on the SGP and TIP implementation published by the Science and Technology Commission of Shanghai Municipality (STC of Shanghai), and "manuals" of the SGP and TIP implementation drafted by the competent departments of the case districts. Most of these formal documents were collected by a robust search of online databases linked to the official websites; the remaining were obtained during field investigation. The second category is internal non-public documents, such as meeting minutes, archives and programme evaluation reports. The authors read, recorded or copied them with the permission of the departments.

The authors' initial interviews with officials and staff in the field took place in face-to-face settings between December 2017 and February 2018; there were three group interviews in the form of meetings held consecutively in the case districts. The

interview format, conducted in a guided, open-ended style, was first submitted to the competent departments in writing; officials who are in charge of and staff who actually undertake the operations of the SGP and TIP were assigned to attend the meeting, and each meeting lasted for approximately two hours. Overall, the authors conducted 16 interviews in which they focused on the issues or problems addressed in the process of operating the SGP and TIP, especially those related to engaging in the MFP. The questions formulated were: "What is the basis for providing matching funds?"; "What is the amount of fund needed every year?"; "How is the relevant work carried out?"; "What comments do you have on this policy?" and "What do you think should be done to improve the policy?". The authors, however, did not adhere to the specific questions, allowing interviewees' responses, wherever necessary, to guide the direction of the interview and for certain sets of questions to be more deeply explored.

FINDINGS

Deconstructing Connotations of Key Factors that Affect Behaviour

As proposition 1 highlights that there are factors that affect the behaviour of governments at both levels, the authors clarify the specific connotations of each key factor based on documents and interviewees' statements.

Engagement benefit for the subordinate government

For the subordinate government, the benefits of engaging in MFP are mainly economic, social and political. First, almost all interviewees mentioned the positive role of matching funds in driving the growth of SMEs in the long run, especially for those at the start-up stage. An official of C district said, "Although the amount of matching funds is not large, the fact that they stand out from the competition for government funding is a recognition of the SMEs' capabilities and helps them to secure financing in the capital market". Interviewees further indicated that "the approved projects are generally in line with the direction of local industry development"; thus the engagement in the MFP indirectly contributes to the local tax base. In fact, past performance evaluations of the MFP had greatly supported this view. However, the benefits took a long time to materialise. The interviewees' general comment was: "The revenue growth has not changed much since we participated in the MFP". As a result, economic and social benefits are not a major consideration in the subordinate government's decision to enthusiastically participate in the MFP.

Second, many interviewees explained that the engagement in the MFP is a matter of being "arranged to do so", such that it is deemed a task which originates as the mandatory requirement from the superior government. Within the Party-state regime, officials of various departments in China's governments at all levels acquire their positions through an appointment system and this can determine how their administrative legitimacy comes from the authorisation from above; therefore, officials always pursue recognition from such superiors by putting in their best performance

in tasks assigned by them. Interviewees commented that "the more active you are in your work, the more favourable will be your superiors' attitude towards you and the more likely you are to get promoted"; the same applies to the engagement in the MFP. This suggests that interests in a political sense, such as promotion, are among the interests pursued by the subordinate government. However, there are no incentive clauses in the MFP policy documents, and interviewees of the districts under study mentioned that "the superior government does not pay much attention to the results of the matching funds implementation". In other words, the results of MFP implementation are not the focus of the superior government when evaluating the subordinate government. Hence to a large extent, whether the subordinate government participates enthusiastically or indolently in terms of the political benefits gained makes little difference.

Engagement cost for the subordinate government

The engagement cost for the subordinate government depends on the latter's responsibilities. Based on the fieldwork, the authors identified three kinds of costs to the subordinate government in engaging in the MFP.

Fiscal opportunity cost

First and most intuitively, for the subordinate government, the engagement in the MFP means the necessity to pay a certain fiscal cost from the public budget for the approved projects. From the interviews, the authors found that the factor of fiscal cost affecting the strategy of the subordinate government was taken in a relative, rather than in an absolute sense. Engagement in the MFP is thus a fiscal opportunity cost but it has different meanings depending on fiscal strength and understanding of the effectiveness of matching funds. On the one hand, the fiscal strength of different subordinate governments varies. In 2017, both Y and C districts accounted for four of the SGPs established in Shanghai. While both districts face nearly identical MFP mandates, the total fiscal revenue of Y district for the year was approximately 95.34 billion yuan, while that of C district was only 41.81 billion yuan. The fiscal pressure faced by the latter was much higher; therefore, the fiscal opportunity cost was relatively greater. On the other hand, there is a different understanding of the effectiveness of matching funds. Interviewees of P district argued that "there are too many enterprises registered in P district being approved by the SGP and TIP; some of them do not have a great potential for growth and are actually not qualified to receive funding and support from the SGP and TIP in our view". Interviewees of P district further stated that "actually, we prefer to spend this part of the budget on fewer enterprises to increase the amount of money that a single enterprise can obtain". As a contrast, interviewees of C district noted that "our annual budget is limited and since these enterprises have already received funding from the municipal government, we could have spent this part of the budget to support other SMEs".

Manpower input cost

Second, for the subordinate government, engaging in the MFP entails not only the provision of certain funds for approved projects, but also the undertaking of a series of management tasks. As shown in Figure 1, with the engagement in the MFP as a link, the district government is involved in the whole process of the operation of SGP and TIP, including the preliminary examination of applications, interim monitoring of projects and final evaluation of programmes. Interviewees expressed the view that "we need to assign someone to take charge of the affairs from beginning to end; it is really a heavy task for us, as we always bear an excessive workload".

Risk event cost

Third, in the process of MFP engagement, subordinate governments may inevitably encounter risks. Interviewees spoke of two kinds of occasional risk events. One is "repeat project establishment". As there are various types of governmental funding programmes managed by different departments, some opportunistic enterprises would apply for different programmes with similar application documents by changing the project name, with the hope of not getting caught. The other is "malicious fraud by the enterprises". Tempted by free governmental funds, a few enterprises with bad intentions chose to cooperate with intermediary agencies and so to capitalise on the former's advantage to increase the probability of approval, even by way of fraud. Since these enterprises have no intention of operating but merely want to cheat in obtaining subsidies from the government, they may disappear before the government can conduct an inspection. Although the superior government makes the final decision on the approval of projects, interviewees complained that "once the risk event occurs, the superior government would blame us for not checking well beforehand". Therefore, in order to prevent depreciation of assets, the subordinate government will monitor the business more carefully when it engages in the MFP enthusiastically rather than indolently, and the cost is correspondingly higher.

Punishment by the superior government

Punishment in this case refers to punitive measures meted out by the superior government for the indolent engagement behaviour of the subordinate government. As the authors browsed through related documents, they did not find any term that was clearly associated with such a penalty. According to the officials of subordinate governments whom the authors interviewed, if their "indolent engagement" were spot-checked by the superior government, they would be obliged to make up the matching funds.

Supervision cost to the superior government

Supervision cost refers to the cost expenditure to the superior government to ensure that the subordinate government can fulfil the policy-based responsibilities of providing both matching funds and the corresponding management. Specifically, supervision

includes checking the availability of matching funds from the subordinate government, monitoring the process and results of project evaluation, and the use of project funds. Indeed, the supervision cost to the municipal government is significantly increased by natural information asymmetry.

Differential Vertical Intergovernmental Interactions

This section illustrates the specific scenario of vertical intergovernmental interaction in the implementation of the MFP in each case district, and proposes the corresponding explanation based on propositions 2 and 3 as outlined above.

Shanghai municipal government: weak supervision

Referring to the supervision behaviour of Shanghai municipal government, interviewees from the three case districts mentioned that "although the MFP requirement in the implementation of the SGP and TIP comes from the uniform regulation of the municipal government, there is no strict measure to supervise it and no relevant accountability rule". The district governments need to submit only an annual summary, which the municipal government usually does not check. In general, the municipal government cares only about whether the matching funds are in place and whether the work it has arranged is completed, and not the district government's work attitude and the quality of work outcomes.

The municipal government's behaviour can be explained by taking proposition 3 into account. On the one hand, due to the ambiguous definitions of enthusiastic or indolent behaviour, no substantive punishment is meted out for the indolent behaviour of the district government; on the other, the supervision cost to the municipal government is significantly increased by natural information asymmetry. Obviously, for the Shanghai municipal government, the fines it can receive from the subordinate government are lower than the supervision cost, hence its optimal strategy is "weak supervision".

Government of Y district: enthusiastic engagement

The authors' interviews in Y district revealed that participation in the MFP plays a positive role in the development of the subordinate government. These interviewees remarked that "we value the innovative SMEs as an accelerator to promote regional industrial transformation. MFP can help us achieve our goal of promoting innovation in local SMEs, so we have been actively promoting it since it was first proposed". The implementation of the SGP and TIP provides a selection mechanism, through which competitive and competent SMEs can be given due recognition. Y district is willing to provide matching funds to these enterprises that have been subsidised by the municipal government for the following reasons. First, an adequate annual budget for S&T affairs ensures that there is no pressure on the expenditure of matching funds. Second, a moderate number of annual approved projects ensures that the Y district

government's supervision workload is not heavy. Third, the government's consistent engagement between strategical priorities and assignment to support SMEs enhances the sense of mission and accomplishment of the officials and staff. Fourth, the preferential treatment for these enterprises is expected to be a demonstration to other SMEs.

Integrating the statements of case Y interviewees with proposition 2 clearly demonstrates that the net benefit for the district from enthusiastically engaging in the MFP is greater than that from indolently engaging in it. This prompted Y district to choose "enthusiastic engagement" as the optimal strategy, regardless of whether the superior government supervises. As a proof of such positive behaviour, interviewees emphasised that "in addition to guaranteeing that enough matching funds are put in place in time, we also provide a series of attractive matching services, including financing facilitation, free training and information dissemination".

Government of P district: indolent engagement I

P district has the strongest fiscal capacity in the city and does not encounter difficulty in providing matching funds itself. The authors learned from interviewees of the district that their main complaint was excessive workload. P district's strong foundation and agglomeration advantage for innovation and entrepreneurship have drawn SMEs to register and develop their business in P district. In the face of coexisting heavy workload and understaffing, two interviewed staff members who directly undertake related tasks complained that "our office is too small to store so many paper documents", and "work seems endless; each of us is responsible for the management of hundreds of projects and we have to deal with many other things at the same time". Interviewees further indicated that "during the busy season, such as when accepting application submissions, we have to resort to the secondment of personnel to help complete the trivial paperwork".

Integrating the arguments of case P interviewees with proposition 2, for P district the manpower input cost of enthusiastically engaging in the MFP is clearly greater than that of indolently engaging in the MFP, and the cost of supervising the enterprises due to enthusiastically engaging in the MFP will also far exceed the cost of indolently engaging due to understaffing, so the net benefit of enthusiastically engaging is less than the net benefit of indolently engaging. As the supervision cost borne by the municipal government is far greater than its punishment of the district government, the optimal strategy for P district is, consequently, indolent engagement. To distinguish this case of indolent engagement in P district from the next case, in C district, discussed below, the authors term the case of P district "indolent engagement I". P district interviewees succinctly summed up their "indolent engagement" as follows: "coping with the required work has been difficult for us, not to mention considering other things, and we really have no energy to improve relevant work".

Government of C district: indolent engagement II

For C district, the heaviest burden from the MFP requirement in the implementation of the SGP and TIP is fiscal spending. On the one hand, as mentioned earlier, with the same matching mandate, C district faces greater pressure to pay for matching funds than Y district does because the former is financially weaker. On the other hand, certain expenditure requirements of the government of C district do not match the willingness to pay, due to the constraints of the fiscal budget. In financial terms, C district's overall financial situation is relatively weak and S&T is not the district's key area of fiscal expenditure. While in terms of the overall situation of SMEs, compared with P district where hundreds of enterprises are approved by the SGP and TIP each year, only a few companies in C district won this honour. C district's interviewees highlighted their observation on the ground: "we are more concerned about how to help SMEs that cannot obtain grants from the municipal government to survive, sometimes a very small monetary shortage is the last straw to crush them".

Integrating the comments of case C interviewees with proposition 2, the fiscal opportunity cost for C district of enthusiastically engaging in the MFP is clearly greater than that of indolently engaging in it, so that the net benefit from enthusiastic engagement is less than that from indolent engagement. As the supervision cost to the municipal government is far higher than its punishment of the district government, the optimal strategy with regard to C district is indolent engagement. Since the incentive under this circumstance is different from that in case P, the authors label case C "indolent engagement II". With regard to C district's performance on the ground, interviewees explained that "we are neutral with respect to the recommendation of enterprises for applying for the SGP and TIP"; "the more approved projects there are, the more matching funds we should provide, and the lesser the money we can spend for projects at the district level". Interviewees further indicated that "actually, we prefer to support our district's own projects for enhancing the innovation base of the district in a wider scope".

Sign of indolent engagement: conspiring with companies to avoid paying matching funds

It is worth exploring the fact that the governments of C and P districts have coincidentally adopted a specific strategy of "indolent engagement" by requiring companies to return the matching funds after the project has been accepted. In the words of interviewees from C district, "a few SMEs are eager to get priority in the endorsement of their project applications"; "they therefore prefer to enter into informal agreements with the district government, promising to abandon their right to get matching funds from the latter in exchange for being recommended". In fact, as mentioned in the previous literature, if a subordinate government adopts other strategies such as reporting false projects and duplicating past projects, not only will it not incur fiscal losses, but it will also receive subsidies that the superior government should have paid to the enterprise.

The reasons why subordinate governments adopt such a strategy may be: (i) The application, evaluation, establishment and acceptance of the SGP and TIP in Shanghai are all under the auspices of the Shanghai Science and Technology Commission, while the Science and Technology Commissions at the district or county level are only "responsible for supervising and managing the small giants within their jurisdictions and assisting in solving the difficulties and problems encountered by the enterprises in the process of implementation". The grassroots government cannot decide whether enterprises can apply for projects or not, so it has to allow them to apply and is forced to give them funding. (ii) The government has to attain not only economic development goals, but also livelihood development goals. SGP and TIP have both economic and social attributes, which present an environment that portrays grassroots governments' rich albeit more complex behaviours. If the goal is only to gain economic benefits, grassroots governments will actively adopt strategies such as imitation and misrepresentation of projects to save costs and siphon off the central funds. However, as China attaches great importance to S&T innovation and implementation of policies, grassroots governments have to attain social development goals such as promoting sustainable development of the local economy, boosting employment and enhancing innovation vitality, and would therefore adopt the strategy of requiring reporting enterprises to commit to returning matching funds to the grassroots government—a tactic that could save costs without hindering the development of local emerging technology enterprises. (iii) Despite the weaker oversight of the superior government, grassroots governments could face high penalties if they discover falsified projects. In the case of collusion with enterprises, grassroots governments face less pressure to supervise. (iv) Due to information asymmetry, it is difficult for local governments to understand the actual situation of each company and disbursing matching funds to all companies as required by the policy may lead to local fiscal loss and failure to reap the expected social benefits. As a result, grassroots governments will avoid political and social risks, and pass on economic risks to the superior government through symbolic means of governance, such as collusion with enterprises.

Political Interest–Fiscal Opportunity Cost–Manpower Input Cost Model

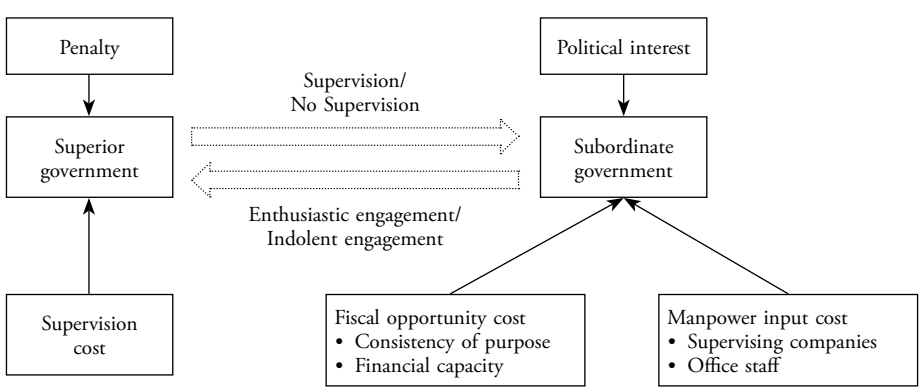
Based on the analysis of the aforementioned cases, the authors revise the initial model constructed based on evolutionary game theory so that the optimised model can explain the gaming behaviour of superior and subordinate governments in the MFP that was designed to promote innovation in SMEs.

The authors derive more specific conclusions, focusing their research on the technology and innovation category of the MFP. Firstly, investment in technological innovation in SMEs does not pay off immediately; technological innovation is a long and complex process with great uncertainty, and it remains uncertain that more capital and manpower invested would result in higher returns. In the game between the superior and subordinate governments, the former considers the long-term benefits, while the latter, due to their term limits, can consider only the immediate benefits.

Therefore, in studying the gaming behaviour of the subordinate government during its term (five years in China), there are zero or negligible economic and social benefits gained, regardless of whether a subordinate government is enthusiastically or indolently engaging. As a result, political benefits take dominance. If the political benefits of enthusiastic engagement are greater than the opportunity cost of capital and the costs of supervising the companies, the evolutionary stable strategies for the subordinate government and superior government are "enthusiastic engagement" and "not supervising", respectively, regardless of whether the fines that the superior government obtains by punishing the subordinate government are higher than its supervision cost. Apparently, "enthusiastic engagement" by the subordinate government and "not supervising" by the superior government represent the ideal situation. However, when the political benefits of active participation are less than the costs of capital and supervision of the companies, a subordinate government will find fewer net benefits to enthusiastically engage than to indolently engage in MFP. At this juncture, the system evolves into the "indolent engagement" and "not supervising" situation if the penalties imposed by the superior government on the subordinate government are less than the costs of supervision. If the penalties of the superior government are greater than the cost of monitoring, the strategy of the superior and subordinate governments will depend on whether the net profit of lazy participation by the subordinate government is still greater than the net profit from enthusiastic participation after subtracting the loss incurred through the penalties. The system evolves into a situation of "indolent engagement" and "supervision", if the subordinate government finds that indolent engagement yields greater net profit than enthusiastic engagement after subtracting the loss due to the penalty. On the other hand, there is no evolutionary stable strategy between the superior and subordinate governments if the subordinate government finds that indolent engagement does not yield greater net profit than enthusiastic engagement after subtracting the loss due to the penalty.

The authors construct an optimisation model (Figure 3) based on the analysis. They also formulate propositions 4, 5 and 6 based on the political interest–fiscal opportunity cost–manpower input cost model.

Figure 3. Political Interest–Fiscal Opportunity Cost–Manpower Input Cost Model



Proposition 4

For MFPs designed to promote SMEs' S&T innovation, the political benefits of participating in the MFP, the fiscal opportunity cost of meeting the matching requirements of superior government, the costs of regulating companies, and the penalties imposed by superior government on subordinate government for indolent engagement are factors that impact the engagement behaviour of the subordinate government; while the supervision cost, the engagement behaviour of subordinate government and the punishment of the subordinate government due to indolent engagement are factors that impact the behaviour of the superior government.

Proposition 5

The behaviour of a subordinate government is always enthusiastic when the political benefits it derives from enthusiastically engaging in the MFP are greater than the sum of the additional capital costs and the costs of regulating the business due to enthusiastic engagement; otherwise, the subordinate government is likely to show initiative only if the penalty is greater than the cost of monitoring by the superior government and the net profit of indolent engagement by the subordinate government is less than the net profit of enthusiastic engagement after subtracting the loss due to the penalty.

Proposition 6

For the superior government, only if the punishment is higher than the supervision cost and the net benefit to the subordinate government from enthusiastically engaging in MFP is less than that from indolently engaging, would it supervise the subordinate government; otherwise, it would not supervise.

CONCLUSIONS AND RECOMMENDATIONS

Innovation capability has become a key determinant of a country's or a region's core competitiveness and governments around the world have prioritised stimulating innovation via policy implementation. Within the general policy framework, the provision of financial grants plays a significant role in promoting growth of SMEs and vigorous innovation activities. In the context of advocating intergovernmental cooperation, vertical intergovernmental interaction plays a critical role in policy effectiveness, including fund and administrative resource efficiency. Given the relatively unique institutional design of the MFP in China that receives scant attention in academia and research, the authors attempt to delineate the MFP implementation to study the interaction mechanism between the superior and subordinate governments in innovation support for SMEs.

The authors combine contingent theoretical derivations with realistic case studies. In the theoretical part, the authors apply evolutionary game theory to demonstrate strategies of both subordinate and superior governments under different circumstances.

Building on that, they formulate several propositions about factors that impact the supervision behaviour of the superior government and the engagement behaviour of the subordinate government, as well as the conditions for the formation of various types of behaviour. In the empirical part, adopting a contrasting perspective, the authors conduct an extensive and comprehensive investigation of three districts in Shanghai city that are characterised by different key endowments. The findings constitute a detailed picture in which differentiated vertical intergovernmental interactions are shaped mainly by the political interest of subordinate governments in participating in the MFP, the fiscal cost of meeting the matching requirements, the cost of regulating companies and the monitoring costs to and penalties exacted by the superior government. The authors label such interaction the political interest–fiscal opportunity cost–manpower input cost model. In short, the superior government always chooses weak supervision, while the subordinate government deliberately chooses to engage enthusiastically or indolently in the MFP.

The authors have further identified both a new form of project alienation in which subordinate governments conspire with enterprises to avoid providing matching funds and the causes of the alienation. The possible causes are: weak management authority; economic and social interests; low penalty levels and challenges of monitoring; and risk transfer and self-protection in the context of corporate lobbying pressure and information asymmetry issues.

The MFP implementation in Y district offers an ideal scenario in which the superior government engages in the MFP enthusiastically and does not need to supervise. However, as different regions are characterised by different endowments, the outcomes of the same policy implemented by a common superior government could be quite different through the vertical intergovernmental interactions. The authors argue that taking a region-based approach should be the main direction for improving the MFP in the innovation support for SMEs and they suggest possible measures to improve it.

First, a threshold of standards of project selection based on local conditions should be formulated. For regions where there is a high concentration of competitive enterprises, the standard of project selection should be raised correspondingly, while for regions where competitive enterprises are lacking, the standard of project selection should be lowered appropriately.

Second, the subordinate government should be bestowed with the autonomy to determine the ratio of matching funds. Such a measure is crucial to regions facing fiscal stress. Regions can exercise their autonomy to lower the ratio according to the fiscal situation. Regions, on the one hand, can reduce the fiscal opportunity cost of engaging in the MFP, thereby increasing the likelihood of enthusiastic engagement; on the other, such autonomy increases their initiative to recommend enterprises to the superior government. Through such comprehensive use alongside the first measure, the improvement of innovation capacity of SMEs in such regions will be accelerated.

Third, a credit and risk management system for SMEs should be established. Since the cost of risk is an indispensable factor affecting the engagement behaviour

of subordinate governments, a sound credit and risk management system for SMEs not only helps the subordinate government to capture in time enterprises that may result in risk events, but also improves the efficiency of the project tracking management.

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